Improving Forecast Accuracy through Data Visualization, Time Series Analysis and Anomaly Detection

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Abstract

In this project, we worked on a real-world demand forecasting problem using a large dataset containing 1,048,575 records. The objective was to understand demand patterns and build forecasting models that can support better decision-making in supply chain and inventory management. During the initial stages, we identified several data quality issues, including 11,239 missing date values and 5,997 negative demand entries, which were removed to ensure accuracy. After cleaning the data and focusing on the period from 2015 to 2019, resulting in 1,000,464 records, we performed detailed exploratory data analysis (EDA) followed by time series modeling. A major anomaly observed in January 2019 significantly affected model performance, but once handled, the RMSE improved from approximately 14.12 million to 6.49 million, representing nearly a 50 percent improvement. The Weighted Moving Average with a three-month window emerged as the best-performing model, demonstrating that simple methods can be highly effective when supported by clean and well-understood data.

Introduction

Demand forecasting is a critical part of modern business operations, especially in supply chain and inventory management. Companies rely on accurate predictions to ensure they have the right amount of stock at the right time. Overestimating demand can lead to excess inventory and increased storage costs, while underestimating demand can result in stockouts and lost sales. In this project, we analyzed a large dataset containing over one million records related to product demand across multiple warehouses and categories. Instead of directly applying complex machine learning models, we focused first on understanding the data through exploratory analysis and then applied simple, interpretable forecasting techniques. This approach ensures that the insights generated are not only accurate but also easy to understand and apply in a real business context.

Data Description

The dataset consists of five key columns representing different aspects of the supply chain, including product identifiers, warehouse locations, product categories, dates, and demand quantities. Initially, most columns were stored as object data types, requiring preprocessing, particularly converting the Date column into datetime format for time-based analysis. The dataset includes 33 product categories, 4 warehouses, and 2,157 unique products. The warehouses are located in Brampton, Oshawa, Surrey, and St John's. During the initial inspection, we found 11,239 missing date values and 5,997 negative demand entries, both of which were removed. Additionally, the year 2014 contained only 640 records compared to over 200,000 records in later years, so it was excluded to maintain consistency. After cleaning, the dataset contained 1,000,464 records covering the period from 2015 to 2019.

Column Name	Initial Data Type	Final Data Type	Description
Product_Code	Object	Object	Unique identifier for each product (2,157 products)
Warehouse	Object	Object	Warehouse location (4 warehouses)
Product_Category	Object	Object	Product grouping (33 categories)
Date	Object	Datetime	Order date for time series analysis
Order_Demand	Float	Float	Quantity demanded

Table 1

Literature Review

Demand forecasting has been a fundamental area of study in operations research, statistics, and data science, as it directly impacts inventory management, supply chain efficiency, and business profitability (Box et al., 2015). Over the years, a wide range of forecasting techniques have been developed, ranging from simple statistical methods to complex machine learning models. However, traditional time series methods remain widely used in practice because they are easy to interpret, computationally efficient, and often provide strong baseline performance (Hyndman & Athanasopoulos, 2018).

One of the most basic and commonly used techniques is the Moving Average method, which works by averaging a fixed number of past observations to predict future values. This method is particularly useful for smoothing short-term fluctuations and identifying overall trends in the

data (Montgomery et al., 2015). However, a key limitation of the Moving Average is that it assigns equal importance to all past observations, which may not always reflect real-world scenarios where recent data is often more relevant.

To address this limitation, the Weighted Moving Average method was introduced, which allows different weights to be assigned to past observations. Typically, more recent observations are given higher weights, making the model more responsive to recent changes in demand. This flexibility often leads to improved forecasting accuracy, especially in datasets where trends evolve over time. In this project, the Weighted Moving Average with a three-month window performed the best, which aligns with findings that shorter windows are more effective in capturing recent trends (Hyndman & Athanasopoulos, 2018).

Another widely used technique is Simple Exponential Smoothing (SES), which also gives more importance to recent observations but does so using an exponential decay factor. Unlike Moving Average methods, SES continuously updates forecasts based on new data, making it more adaptive (Box et al., 2015). However, SES assumes that the data has no strong trend or seasonality, which can limit its effectiveness in more complex datasets. In addition to these methods, regression-based approaches are sometimes used for forecasting, where demand is modeled as a function of time and other variables. These models can capture trends and seasonal effects, but they often struggle when the data contains irregular fluctuations or anomalies, as observed in this project (Montgomery et al., 2015).

More advanced forecasting techniques, such as ARIMA (AutoRegressive Integrated Moving Average) and machine learning models like XGBoost, are capable of capturing complex patterns, including seasonality and non-linear relationships. However, these models require more computational resources, careful parameter tuning, and may lack interpretability. In many real-world business scenarios, simpler models are preferred because they are easier to implement and explain to stakeholders (Hyndman & Athanasopoulos, 2018). Overall, the literature suggests that while advanced models can offer improvements, the effectiveness of any forecasting method depends heavily on data quality and preprocessing. This aligns with the findings of this project, where handling anomalies and cleaning the data led to a significant improvement in model performance, even when using relatively simple forecasting techniques.

Approach

The approach followed a structured pipeline to ensure accuracy and clarity. First, the data was cleaned by removing all invalid entries, including 5,997 negative demand values and 11,239 rows with missing dates. The Date column was converted into datetime format, and invalid entries were removed. The dataset was then filtered to include only the years 2015 to 2019, resulting in 1,000,464 clean records. Next, the data was aggregated into monthly demand values, which helped reduce noise and highlight trends. A critical step in the process was handling the

anomaly observed in January 2019. To evaluate its impact, three versions of the dataset were created: original, anomaly removed, and anomaly imputed using the average of adjacent months. The dataset was then split into training and validation sets using an 80–20 split. Multiple forecasting models, including Naive, Moving Average, Weighted Moving Average, Simple Exponential Smoothing, and Regression, were applied and evaluated using RMSE, MAE, and MAPE.

Exploratory Data Analysis (EDA)

Before building any models, we conducted detailed exploratory data analysis to understand the structure and patterns in the dataset. This step was important because it helped us identify trends, imbalances, and potential issues that could affect forecasting performance.

First, we analyzed the distribution of demand across product categories. As shown in Figure 1, Category 019 dominates the dataset, contributing approximately 84 percent of total demand. The next highest category contributes significantly less, which highlights a strong imbalance in the data. This indicates that most of the business operations are concentrated around a single category, and any forecasting model must capture this dominance accurately.

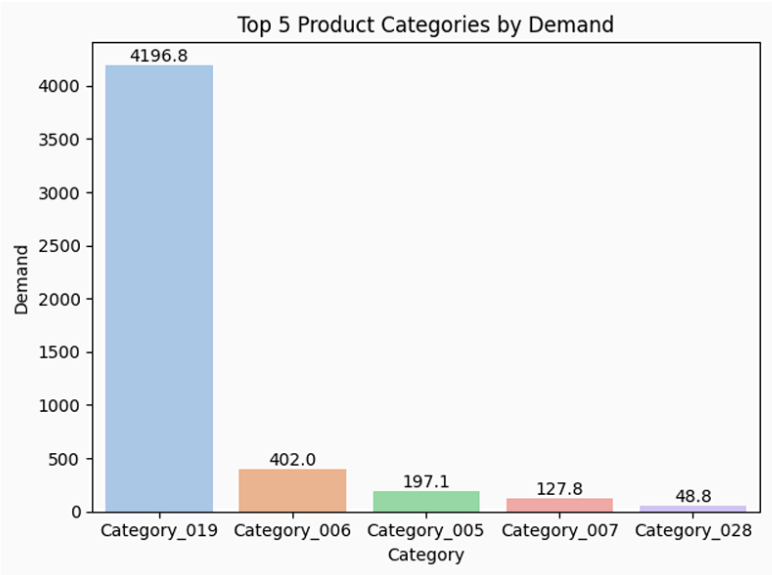

Figure 1

Next, we examined demand distribution across warehouses, as illustrated in Figure 2. The Brampton warehouse accounts for approximately 66 percent of total demand, making it the primary operational hub. In contrast, St John's contributes only about 3 percent. This uneven distribution suggests that forecasting and inventory management strategies should focus heavily

on Brampton, while also considering whether resources in smaller warehouses are being underutilized.

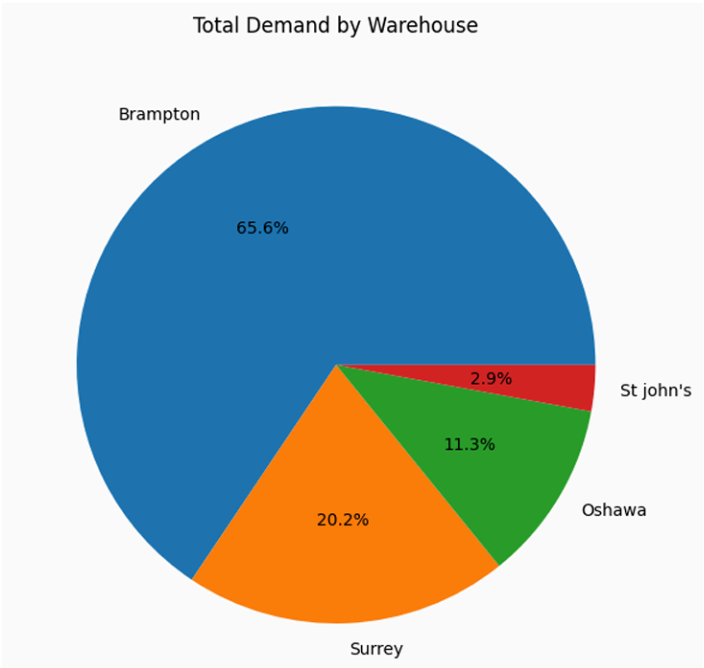

Figure 2

We also explored how demand varies across product categories within each warehouse using a heatmap (Figure 3). The heatmap shows that although Category 019 dominates overall demand, the distribution of other categories varies across warehouses. This indicates that demand patterns are not uniform across locations and reinforces the need for warehouse-specific forecasting.

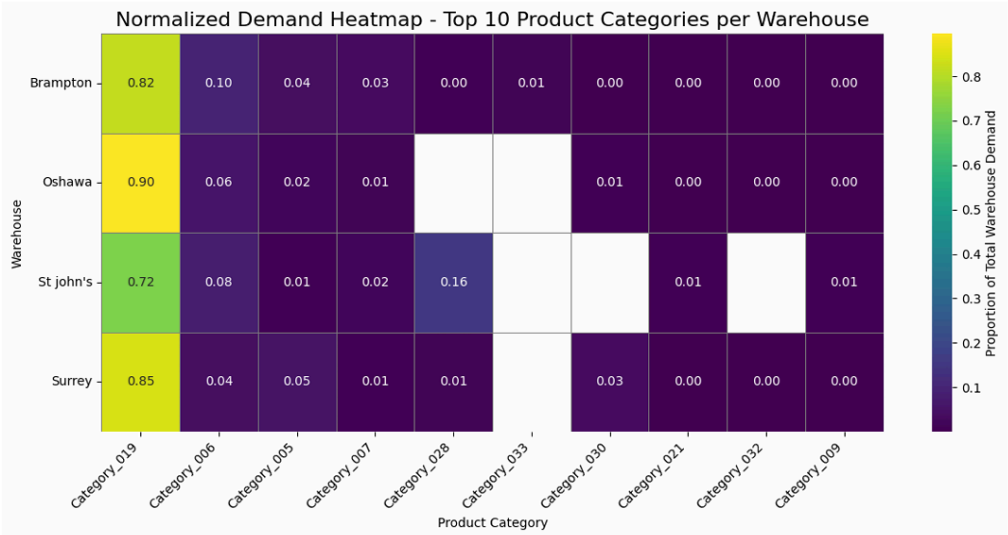

Figure 3

To understand trends over time, we aggregated the data into monthly demand values and plotted them as a time series (Figure 4). The plot shows a steady increase in demand from 2015 to 2018, indicating consistent business growth. However, a sharp drop is observed in January 2019, which is inconsistent with the surrounding data points. This anomaly was clearly visible in the visualization and became a key focus of our analysis.

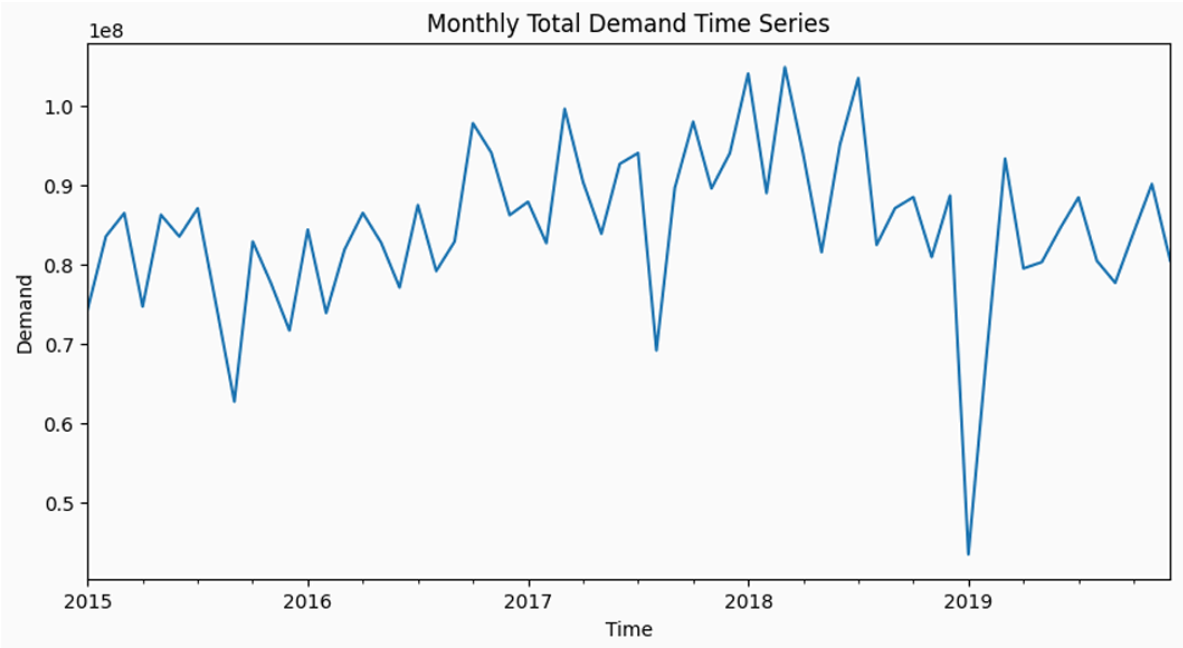

Figure 4

We also analyzed demand trends at the warehouse level and found that Brampton shows relatively stable demand over time, while Oshawa and St John's show slight upward trends, and Surrey shows a slight downward trend. This variation suggests that demand behavior differs across locations and should be modeled separately for better accuracy.

At a more granular level, we examined the top products within each warehouse. For example, Product_1359 in the Brampton warehouse had the highest demand, exceeding 468 million units. However, even high-demand products showed significant fluctuations over time, indicating that demand is not only high but also volatile. This further emphasizes the importance of product-level forecasting.

Overall, the EDA phase helped us understand the structure of the data, identify key patterns, and detect anomalies, all of which guided the modeling process and improved the final results.

Findings and Discussion

The findings from the analysis provide several important insights that are directly relevant to business operations. The dominance of Category 019, which accounts for 84 percent of total demand, suggests that focusing on this category can lead to the greatest improvements in efficiency and profitability. Similarly, the Brampton warehouse, which handles 66 percent of demand, plays a critical role in the supply chain and requires robust forecasting and inventory management systems.

The anomaly observed in January 2019 had a significant impact on model performance. When included, the RMSE was approximately 14.12 million and error rates ranged from 15 to 20 percent. After removing or correcting the anomaly, the RMSE dropped to 6.49 million and error rates reduced to between 8 and 12 percent. This demonstrates that data quality has a direct and substantial impact on forecasting accuracy.

The comparison of models shows that the Weighted Moving Average with a three-month window achieved the best performance, with the lowest RMSE of 6,499,882. The Naive model also performed well, achieving the lowest MAE of 5,146,591 and MAPE of 6.16 percent, indicating that simple models can still be effective. In contrast, models such as Moving Average and Regression performed poorly, with RMSE values exceeding 10 million, suggesting that they were unable to capture the variability in demand effectively.

Model	**RMSE (Original)**	**RMSE (Cleaned)**	**MAE (Cleaned)**	**MAPE (%)**	**Performance**
WMA	14,122,970	6,499,882	5,226,098	6.53	Best Overall
Naive	15,547,440	6,674,875	5,146,591	6.16	Strong (Best MAE & MAPE)
WMA(6)	14,429,210	7,443,762	6,034,482	7.67	Good
SES	15,569,970	8,540,515	6,543,323	8.43	Moderate
MA	17,482,760	11,026,870	9,230,491	11.76	Weak
Regression	17,967,270	10,041,320	8,329,902	10.46	Weak

Table 2

These results highlight that the choice of model is important, but the quality of the data has an even greater impact on performance. From a business perspective, the insights suggest that forecasting should be performed at both warehouse and product levels to capture variability more effectively.

Conclusion

In conclusion, this project demonstrates that effective demand forecasting depends not only on the choice of model but also on the quality of the data and the depth of analysis. By removing invalid values and addressing anomalies, we were able to significantly improve forecasting performance, reducing RMSE by nearly 50 percent. The Weighted Moving Average model with a three-month window emerged as the best-performing method due to its ability to balance recent trends with historical data. The analysis also showed that demand is highly concentrated in Category 019 and the Brampton warehouse, which together account for the majority of business activity. These findings suggest that companies should focus their forecasting and operational strategies on these key areas while also considering more granular forecasting at the warehouse and product levels. Although advanced models such as ARIMA and XGBoost can be explored in future work, this study shows that simple models can deliver strong results when applied to well-prepared data.

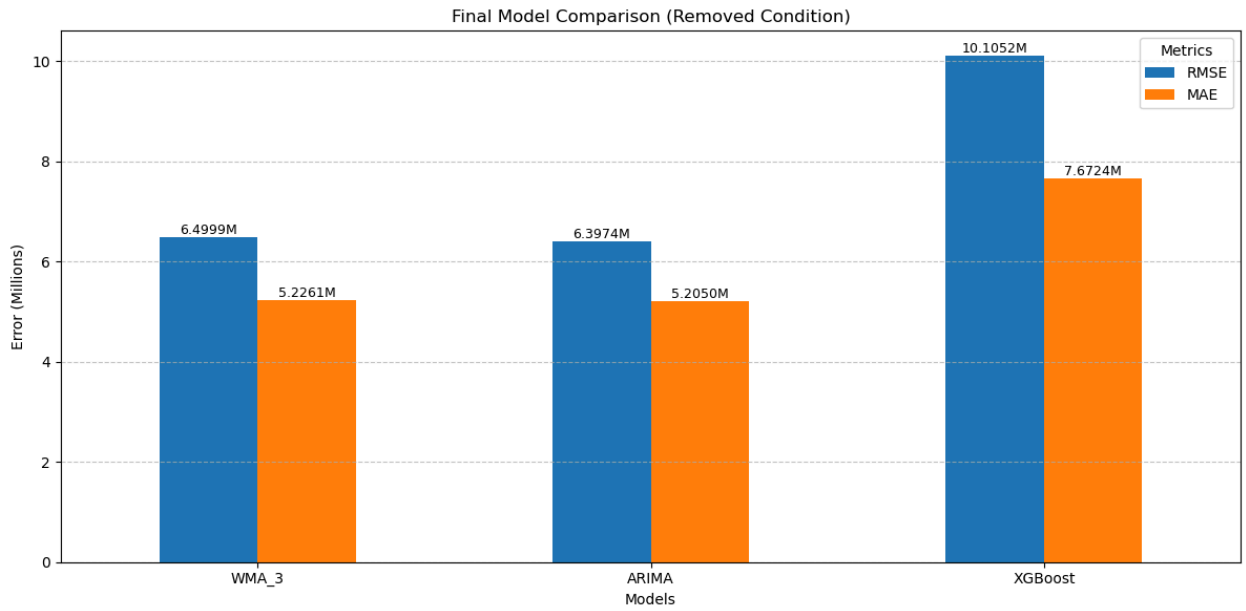

Figure 5

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Who did What Table

No.	Team Member	Role / Contribution	Details
1	**Sam Leslie**	Data Cleaning & Preprocessing	handled data cleaning and preprocessing (fixing missing/invalid data and preparing the dataset).
2	**Yash Pratap Singh**	Modeling, Calculations & Metrics	worked on forecasting models and calculated evaluation metrics (RMSE, MAE, MAPE).
3	**Labdhi Zatakia**	Anomaly Detection & Report Writing	Anomaly Detection & Report Writing, Identified and handled January 2019 anomaly, created corrected datasets, and contributed to report writing.
4	**Banshi Keshwala**	Report Writing & Analysis	led the report writing, analysis, and overall documentation structure.
5	**Mohit Dabas**	Modeling & Presentation	implemented models, managed training/validation, and worked on presentation slides.